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- (a) Read the image **woman.tif** obtaining your Fourier centralized spectrum. Apply both highpass and lowpass butterworth filter, plotting the final image, the 2D and 3D filter visualization.
- (b) Create a **H=highEmphasisFilt(a,b,f, type, show)** function. A high-emphasis filter, given by:

$$H_{HFE} = a + bH_{HP}$$

where a is the offset, b is the multiplier and f the image. Plot the 3D visualization and 2D visualization of this filter. The **type** parameter must be the same in the `lpfilt` function. The **show** parameter must be boolean, if True must show the filter plotting

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(c) Test your **highEnphasisFilt** function with the image **chestXray.tif**. Show the spectrum after and before the filtering processing in (b). Do the **histogram equalization (histEq4e or histEq)** in the filtered image and show the result.

What should be uploaded?

- 1.m file (commented);
- 2.Images obtained, conclusions and comparison about the images and the used tactics.
- 3.Reports.